Fetch My Route!

System Testing Document (STD)

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1. Introduction

1.1. Purpose of This Document

1.1.1. The purpose of this document is to outline the testing process for the UMBC Accessibility GPS website. It will go over the results of our testing and key takeaways from our process. The intended audiences for this document are the software developers, project owner, and our stakeholder. The software developers and project owner will use this to know the scope of our tests, and how they were done. The Stakeholder will use this document to understand our testing for the product.

1.2. References

1.2.1. SRS Document - UMBC Accessibility GPS

1.2.2. UMBC Accessibility Routes Map

2. Testing Process

2.1. Description

2.1.1. Our testing process focussed mainly on both unit and system testing. We wanted to effectively test every component as it was developed, which would allow us to then test the entire system. Each member of our team would extensively test each feature implementation to improve the efficiency of our team when we began system testing.

2.2. Testing Sessions

Date	Location	Time Started	Time Ended	Testers	Use Case
11/18/25	Remote	8:15pm	9.00pm	Aidan Denham, Alex Marbut, Celestine Sumah, Ethan Michalik	Get outdoor directions, Login/Logout, Get indoor directions, Provide feedback, Save route, View saved routes, Load route, Change accessibility settings
11/20/25	Remote	7:00pm	8:15pm	Aidan Denham,	User Authentication

				Alex Marbut, Celestine Sumah, Ethan Michalik	User Authentication User Authentication User Authentication
11/23/25	Remote	4:00pm	6:00pm	Aidan Denham, Alex Marbut, Celestine Sumah, Ethan Michalik	Draw Map Route Draw Map Route Draw Map Route Draw Map Route
11/25/25	In Person	8:15pm	9:00pm	Aidan Denham, Alex Marbut, Celestine Sumah, Ethan Michalik	Load Leaflet Map Load Leaflet Map Load Leaflet Map Load Leaflet Map
11/29/25	Remote	10:00pm	11:30pm	Aidan Denham, Alex Marbut, Celestine Sumah, Ethan Michalik	Load Building Directory Load Building Directory Load Building Directory Load Building Directory
11/30/25	Remote	10:00pm	11:30pm	Aidan Denham, Alex Marbut, Celestine Sumah, Ethan Michalik	Enable Map Click Enable Map Click Enable Map Click Enable Map Click

2.3. Impressions of the Process

2.3.1. Our testing process was very effective. Going through the user flow of our processes, we were able to effectively root out any bugs the user may run into. Given the functional architecture of our system, any bugs we ran into were not system breaking, and we were able to single them out one by one. Before testing our website was subpar but functional. After testing our website was in a much better position, and had little to no interference on the user.

Our best modular unit is the login/logout functionality. It is a very small module and thus doesn't run into many issues. Testing was very straightforward as we mainly needed functionality for the user to sign in with their UMBC email via google. This way google would automatically handle the process of authenticating the user, since UMBC has that systemic functionality already. After getting over the hurdle of getting google authentication to work, this module has the smallest chance for bugs, as most of the functionality is handled by a third-party source.

Our worst module unit is our indoor directions functionality. This module by far contains the most chances for bugs to appear due to all the parts of

our system related to it. We have to give autofill suggestions for building locations, ensure the inputted location is a real location, ensure the coordinates for building match up with their location on the map, ensure the paths follow our given routes, and ensure our routes are accurate to accessible paths around campus. A lot of data was gathered to ensure all of our routes are accessible, but with the size of campus it's always possible some oversights were made. The rest of these parts are fairly minor alone, but together ultimately provide the most spots for bugs to appear and be missed. Through our testing we believe to have ironed out most of the bugs, but this module has the highest chance of new ones occurring.

3. Test Results

3.1. Testing Suite

Test Case ID	Test Case Description	Preconditions	Test Steps	Expected Results	Actual Result	Status
Get Outdoor Directions						
TC01	User can find routes from one building to another	N/A	1. Enter building into source destination 2. Enter building into target destination 3. Verify calculated route	Map shows an accurate route between the two buildings	Map shows accurate route	Pass
User Login/Logout						
TC02	User can sign in with their UBMC email	User has a UMBC email	1. Click login button 2. Type in student email and password 3. Google authenticates student account	User successfully logs in with their student email	User successfully logged in	Pass
TC03	User can log out of their account	User is logged in	1. Click the account button 2. Click Log out	User successfully logs out of the website	User successfully logged out	Pass

Get Indoor Directions						
TC04	User can find routes through buildings	User is logged in	1. Enter building into source destination 2. Enter building into target destination 3. Verify calculated route	Calculated route can go through buildings and can utilize elevators	Route correctly paths through buildings	Pass
Provide Feedback						
TC05	User can submit feedback on the website	User is logged in	1. Click feedback button 2.Type in feedback into the text field 3. Click submit button	User feedback is sent as email to the admin	Admin received user feedback	Pass
Save Route						
TC06	User can save a route to load in the future	User is logged in, User has filled out the destination input fields	1. Click the save route button	Route is saved and can be viewed in their account	Logged in user can view saved routes	Pass
View Saved Routes						
TC07	User can view their saved routes	User is logged in	1. Click the account button 2. Click view saved routes	A list of all saved routes is shown to the user	User can view saved routes	Pass
TC08	User can load a saved route	User is logged in	1. View saved routes 2. Click a saved route 3. Click the load button	The saved route is loaded into the input fields	The saved route is loaded into the input fields	Pass
Change Accessibility Settings						
TC09	User can turn on high contrast mode	N/A	1. Click accessibility features button 2. Click high	Website's colors change to high contrast colors	Website changes to use high contrast colors	Pass

			contrast button			
TC10	User can increase font size	N/A	1. Click accessibility feature button 2. Click plus (+) button to increase font size	Font size increases	Font size increases	Pass
TC11	User can decrease font size	N/A	1. Click accessibility feature button 2. Click minus (-) button to decrease font size	Font size decreases	Font size decreases	Pass

3.2. Test Results

Test Case IDs	Tester	Feature Tested	Result	Notes
TC01	Aidan Denham	Get Outdoor Directions	Passed	
TC02 - TC03	Celestine Sumah	Login/logout	Passed	
TC4	Aidan Denham	Get Indoor Directions	Passed	
TC5	Celestine Sumah	Provide Feedback	Passed	
TC6	Alex Marbut	Save Route	Passed	
TC07 - TC08	Alex Marbut	View Saved Route	Passed	
TC09 - TC11	Ethan Micihalik	Change Accessibility Settings	Passed	

Appendix A - Peer Review Sign-off

By signing this document, all members of the team have agreed to this document's contents and format. If there are major contentions with the document, members will not sign until they have been discussed.

Name	Signature	Date	Comments
Aidan Denham	Aidan Denham	12/1/2025	N/A
Celestine Sumah	Celestine Sumah	12/1/2025	N/A
Ethan Michalik	Ethan Michalik	12/1/2025	N/A
Alex Marbut	Alex Marbut	12/1/2025	N/A

Appendix B - Document Contributions

Contributions by: Aidan Denham, Alex Marbut, Celestine Sumah, Ethan Michalik